Day 1: Static Routing Concepts

Theory

Q1. What is static routing?

Static routing is a method of configuring routes in a router's routing table manually, by the network administrator, instead of letting the router learn them automatically through a routing protocol. Each static route tells the router exactly which path to take to reach a specific destination network, usually by specifying the destination network address, subnet mask, and the next-hop IP address (or exit interface).

For example, on Router0 (Ahmedabad) in this lab, a static route was added telling it that the 192.168.20.0/24 network (Surat's LAN) is reached via the serial link to 10.0.0.2. Because static routes are fixed, the router will always use that exact path unless the administrator changes the configuration — it does not automatically adjust if the network topology changes.

Q2. Advantages and disadvantages of static routes

Advantages	Disadvantages
Simple to configure in small networks with few routers.	Does not scale well — becomes very hard to manage as the number of routers/networks grows.
Uses no CPU/bandwidth for route calculation (no routing protocol overhead).	Does not automatically adapt to link failures or topology changes; needs manual reconfiguration.
More secure — no route advertisements are sent, so there is nothing for an attacker to intercept or spoof.	Administrator must manually update every router when the network changes, which is time-consuming and error-prone.
Predictable and fully controlled path selection, useful for stub networks.	No automatic failover/redundancy — if the configured path goes down, traffic simply stops unless a backup route is manually configured.

Practical

Task 1: Two-router topology with a serial link

A simple network was built in Cisco Packet Tracer with two 2901 routers connected via a serial cable (Se0/0/0 on each side). Each router also connects to its own LAN through a 2960-24TT switch and a PC. IP addressing used:

- Router0 (Ahmedabad): Se0/0/0 = 10.0.0.1/30, GigabitEthernet0/0 = 192.168.10.1/24
- Router1 (Surat): Se0/0/0 = 10.0.0.2/30, GigabitEthernet0/0 = 192.168.20.1/24
- PC0 (Ahmedabad LAN): 192.168.10.2/24, Gateway 192.168.10.1
- PC1 (Surat LAN): 192.168.20.2/24, Gateway 192.168.20.1

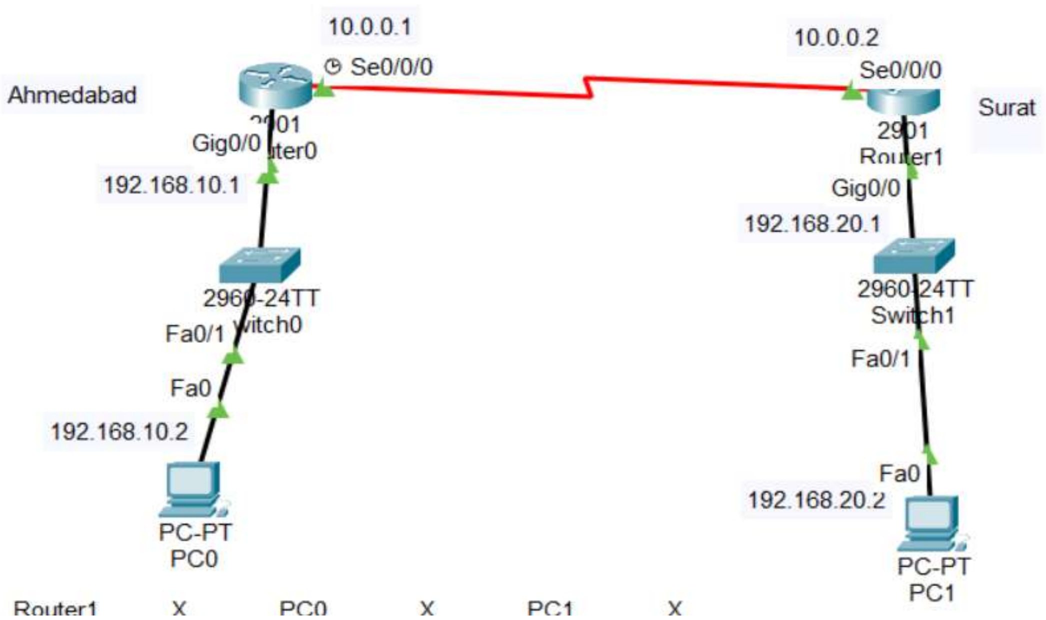

Figure 1: Two-router topology — Ahmedabad (Router0) and Surat (Router1) connected via serial link

Task 2: Static route from Router1 to PC on Router0's LAN, tested with ping

Router0 (Ahmedabad) configuration:

```
Router0> enable
Router0# configure terminal
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.10.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface Serial0/0/0
Router0(config-if)# ip address 10.0.0.1 255.255.255.252
Router0(config-if)# clock rate 64000
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

Router1 (Surat) configuration:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# ip address 192.168.20.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Serial0/0/0
Router1(config-if)# ip address 10.0.0.2 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

With the static route ip route 192.168.20.0 255.255.255.0 10.0.0.2 added on Router0 (and the reverse route on Router1), Router0 knows to forward any traffic for the 192.168.20.0/24 network out through its serial interface to next-hop 10.0.0.2. Connectivity was verified by pinging from each PC to the PC on the other router's LAN, using the PC command prompt.

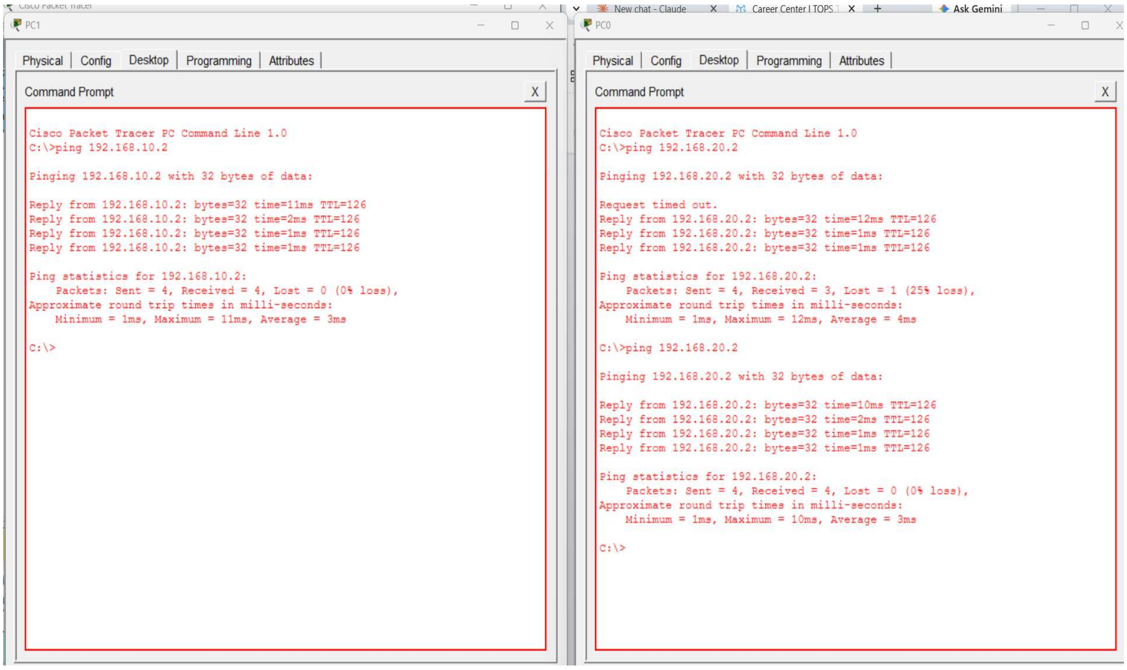

Figure 2: Ping test results — PC1 → 192.168.10.2 (left) and PC0 → 192.168.20.2 (right)

PC1 (Surat) pinging 192.168.10.2 (PC0 on Ahmedabad's LAN) succeeded with all 4 packets received (0% loss). PC0 (Ahmedabad) pinging 192.168.20.2 (PC1 on Surat's LAN) showed 1 timeout on the very first ping attempt (25% loss) because the router had to first resolve the ARP entry for the destination — a second ping to the same address then succeeded with all 4 packets received (0% loss). This confirms that the static routes on both routers are working correctly and end-to-end connectivity exists between the two LANs across the serial link.

Task 3: Zomato-style delivery network simulation (Ahmedabad ↔ Surat)

Router0 was treated as the Ahmedabad branch and Router1 as the Surat branch, each representing a city where a Zomato-style delivery service operates. Devices on each branch's LAN (e.g., PC0 in Ahmedabad, PC1 in Surat) represent order-processing systems or delivery-tracking terminals local to that city.

The static routes configured in Task 2 (ip route 192.168.20.0 255.255.255.0 10.0.0.2 on Router0, and ip route 192.168.10.0 255.255.255.0 10.0.0.1 on Router1) are exactly what allows a device in the Ahmedabad LAN to reach a device in the Surat LAN and vice versa — the same topology and configuration used in Task 2 doubles as this Zomato-style branch-to-branch simulation. The successful pings shown in Figure 2 demonstrate that an order or status update originating in one city's LAN can successfully reach the other city's LAN through the two routers, just as an update from an Ahmedabad restaurant's system would need to reach a Surat delivery partner's tracking system (or a central server) over a WAN link.

Task 4: Advantages and disadvantages of static routing vs. dynamic routing (using IRCTC / Swiggy examples)

Static routing means the admin manually tells the router which path to use. Dynamic routing means the routers talk to each other and figure out the paths on their own. Here is how that plays out for apps like IRCTC and Swiggy:

Point	Explanation	Example (IRCTC / Swiggy)
Advantage 1: Fixed and secure path	The path never changes, and the router does not broadcast route info, so it's harder for an attacker to interfere.	IRCTC's booking servers and payment servers could connect over a static route, since that link rarely changes and needs to stay secure.
Advantage 2: Less router work	The router doesn't spend CPU or bandwidth constantly exchanging route updates.	A fixed link between Swiggy's order database and payment gateway can use a static route, keeping order confirmation fast.
Disadvantage 1: Doesn't scale well	Every time a new location is added, the admin must manually add a route on every router.	If Swiggy opens in a new city, someone has to manually configure new routes everywhere — dynamic routing would learn this automatically.
Disadvantage 2: No auto-recovery from failure	If the configured link goes down, the router cannot find another way by itself. Someone has to fix it manually.	If a link between two IRCTC data centers fails during Tatkal booking (a peak-traffic period), bookings fail until an admin manually adds a new route.

In short: static routing is fine for small, stable, or sensitive parts of the network, but large and constantly changing networks like IRCTC or Swiggy mostly need dynamic routing instead.